how the implosion could be simulated on a computer, including all the key effects that were so important for the hydrogen bomb: pressure, nuclear reactions, shock waves, heat, radiation, mass ejection.

Stimulated by these discussions, Imshennik and Nadezhin simulated the implosion of stars with small mass—and verified, independently of Colgate and White in America, Zwicky's conjectures about supernovae. In parallel, Podurets simulated the implosion of a massive star. Podurets's results, published almost simultaneously with those from May and White in America, were nearly identical to the Americans'. There could be no doubt. Implosion produces black holes, and does so in just the way that Oppenheimer and Snyder had claimed.

The adaptation of bomb design codes to simulate stellar implosion is just one of many intimate connections between nuclear weapons and astrophysics. These connections were obvious to Sakharov in 1948. Upon being ordered to join Tamm's bomb design team, he embarked on a study of astrophysics to prepare himself. My own nose was rubbed into the connections unexpectedly in 1969.

I never really wanted to know what the Teller-Ulam/Sakharov–Zel'dovich idea was. The superbomb, one that by virtue of their idea could "be arbitrarily powerful," seemed obscene to me, and I didn't want even to speculate about how it worked. But my quest to understand the possible roles of neutron stars in the Universe forced the Teller–Ulam idea onto my consciousness.

Zel'dovich, several years earlier, had pointed out that gas from interstellar space or a nearby star, falling onto a neutron star, should heat up and shine brightly: It should become so hot, in fact, that it radiates mostly high-energy X-rays rather than less energetic light. The infalling gas controls the rate of outflow of X-rays, Zel'dovich argued, and conversely, the outflowing X-rays control the rate of infall of gas. Thereby, the two, gas and X-rays, working together, produce a steady, *self-regulated flow.* If the gas falls in at too high a rate, then it will produce lots of X-rays, and the outpouring X-rays will strike the infalling gas, producing an outward pressure that slows the gas's fall (Figure 6.4a). On the other hand, if the gas falls in at too low a rate, then it produces so few X-rays that they are powerless to slow the infalling gas, so the infall rate increases. There is just